CrossMind

AI BEAVERS Founder Hackathon | House of AI Hamburg | June 6, 2026

Split inference for sensitive data

- The problem
  In large organizations, local AI models can securely consult centralized specialist LLMs by minimizing sensitive data anywhere in the pipeline.
- The buyer
  Hospital-group IT, clinic compliance, enterprise AI leads: anyone blocking raw data export to a central model.
- Across domains
  Healthcare, finance, logistics, legal, industrial. For this hackathon: clinic to hospital.

DEMO - Research prototype only. Not medical advice, diagnosis, or treatment.

How it works: Our Demo

- **Clinic runs Qwen 2.5 locally**

  Encodes patient text into a numeric vector. The original words never leave the clinic.
- **Hospital runs Llama 3.1**

  Receives only numbers (vectors), not your sentence, not your words, not your document.
- **Linear alignment connects them**

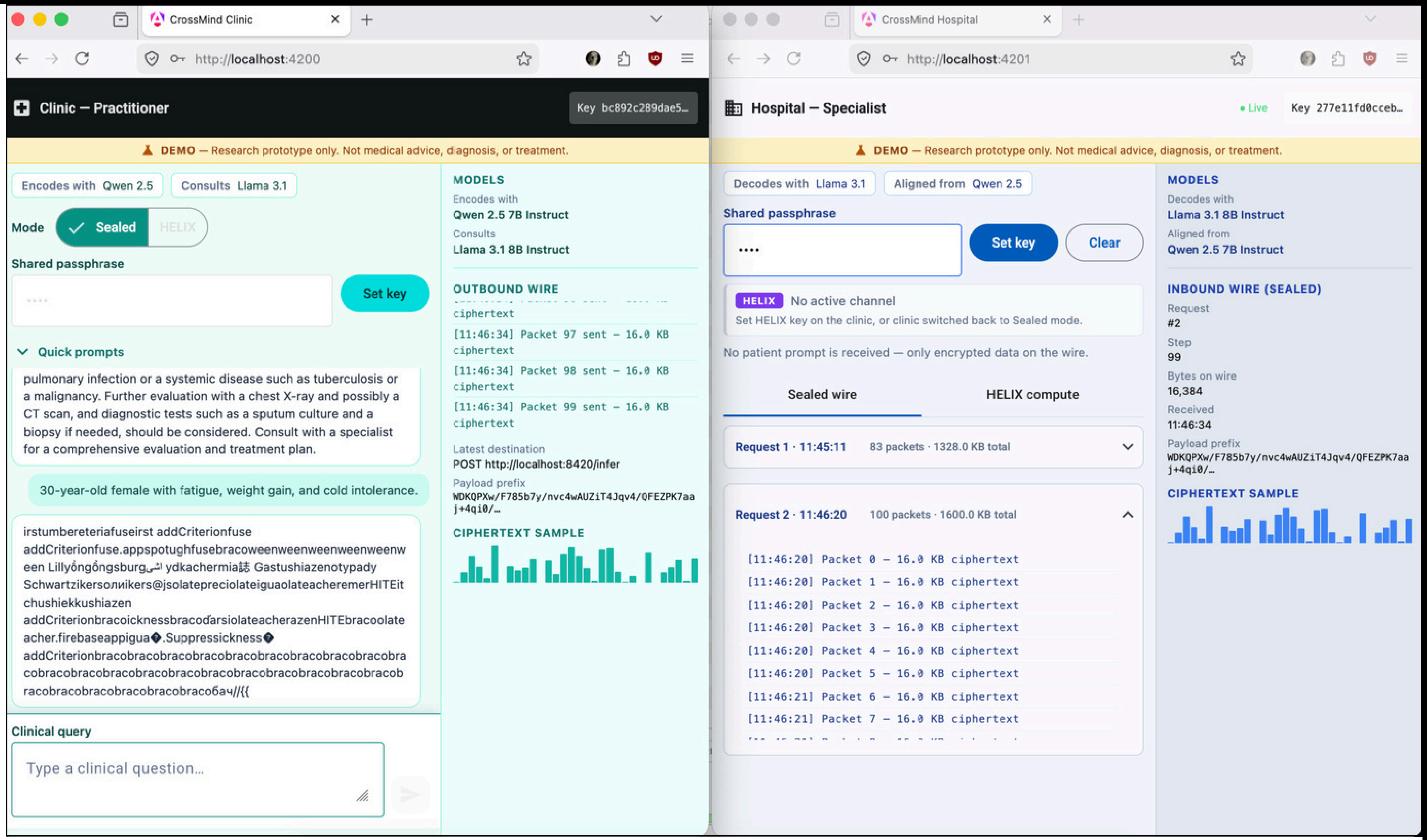
  A trained matrix translates Qwen's internal representation into Llama's space and obfuscates it via rotation (Gorbett & Jana, 2025).
- **Trained on medical Q&A pairs**

  The alignment is optimized for clinical domain accuracy, trained on a 4K q&a dataset leading to an alignment cosine ~0.83.

```
Clinic (Qwen) encodes text into vector  -->  alignment matrix translates it  -->  obfuscated via
passphrase  -->  Hospital (Llama) generates from vector  -->  token back
```

Privacy Layer 1: Sealed (wire obfuscation)

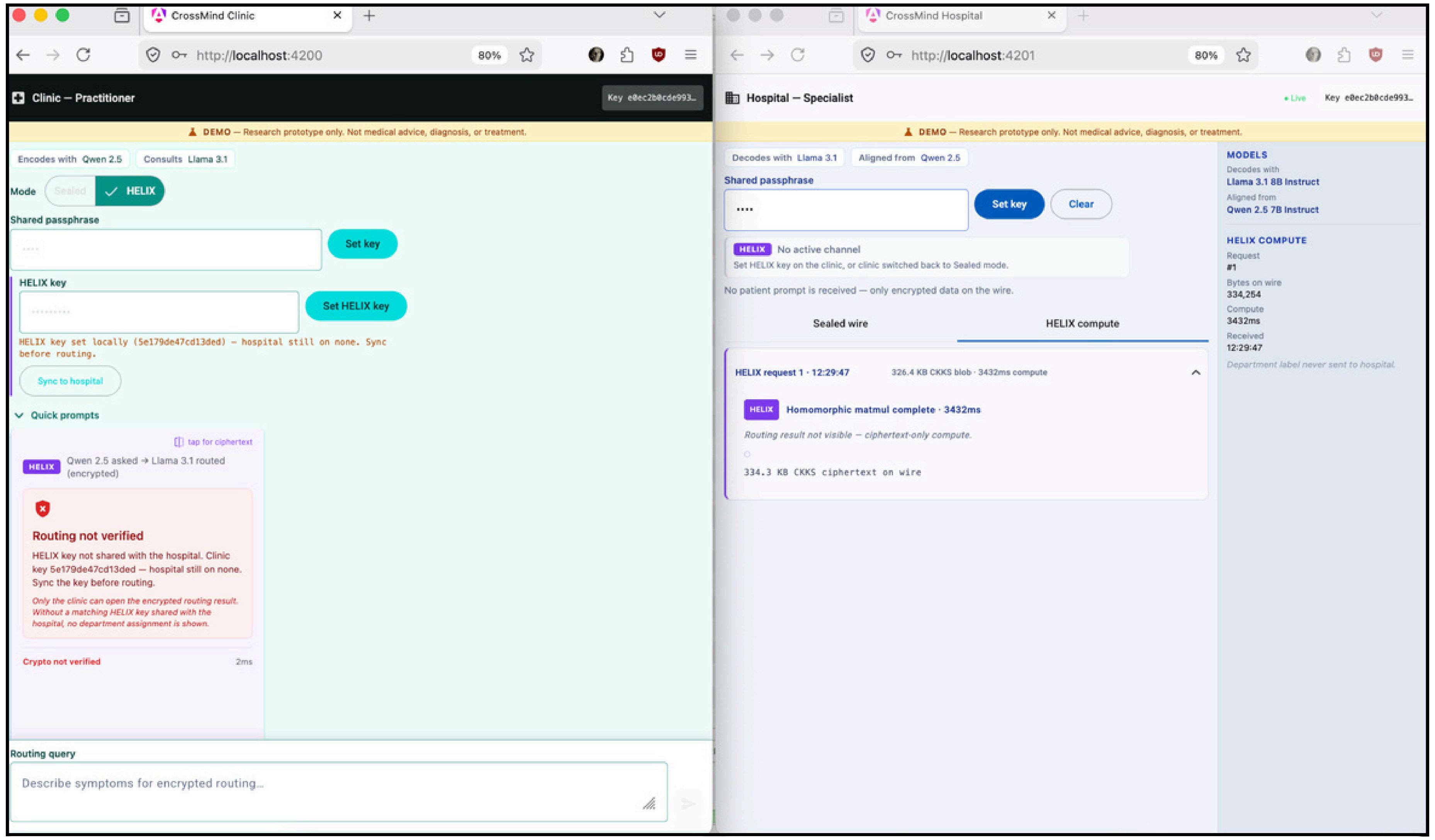
- **Vector-by-vector transmission**
  Each hidden state travels individually. This adds a structural privacy layer: no full document/prompt crosses the wire.
- **Shared passphrase rotates each vector**
  Before sending, the clinic rotates the vector using a shared secret. The specialist only ever sees rotated numbers.
- **Wrong passphrase = garbled nonsense**
  Without the matching key, intercepted data is meaningless. See screenshot: mismatched keys produce noise.
- **Protects against passive eavesdroppers**
  Anyone tapping the network link gets rotated vectors for Obfuscation



Mismatched passphrase: clinic and hospital have different keys.
Output is garbled. Data stays mangled on the wire.

Privacy Layer 2: HELIX (homomorphic encryption)

- **CKKS homomorphic encryption**
  The hospital computes on ciphertext. It never sees the plaintext vector at any point.
- **Department routing (5 classes)**
  Cardiology, Neurology, Oncology, Orthopedics, General Medicine.
- **Hospital never sees the department label**
  Only the clinic can decrypt the routing result. The hospital is truly blind.
- **~3.4 seconds per routing request**
  Homomorphic operations are computationally expensive. Full-vocab generation (128k tokens) is not feasible yet. But a 5-class routing head is practical today.



HELIX routing: crypto verified, department decrypted only by clinic.

DEMO - Research prototype only. Not medical advice, diagnosis, or treatment.

What our privacy layers do and what they don't

What it DOES

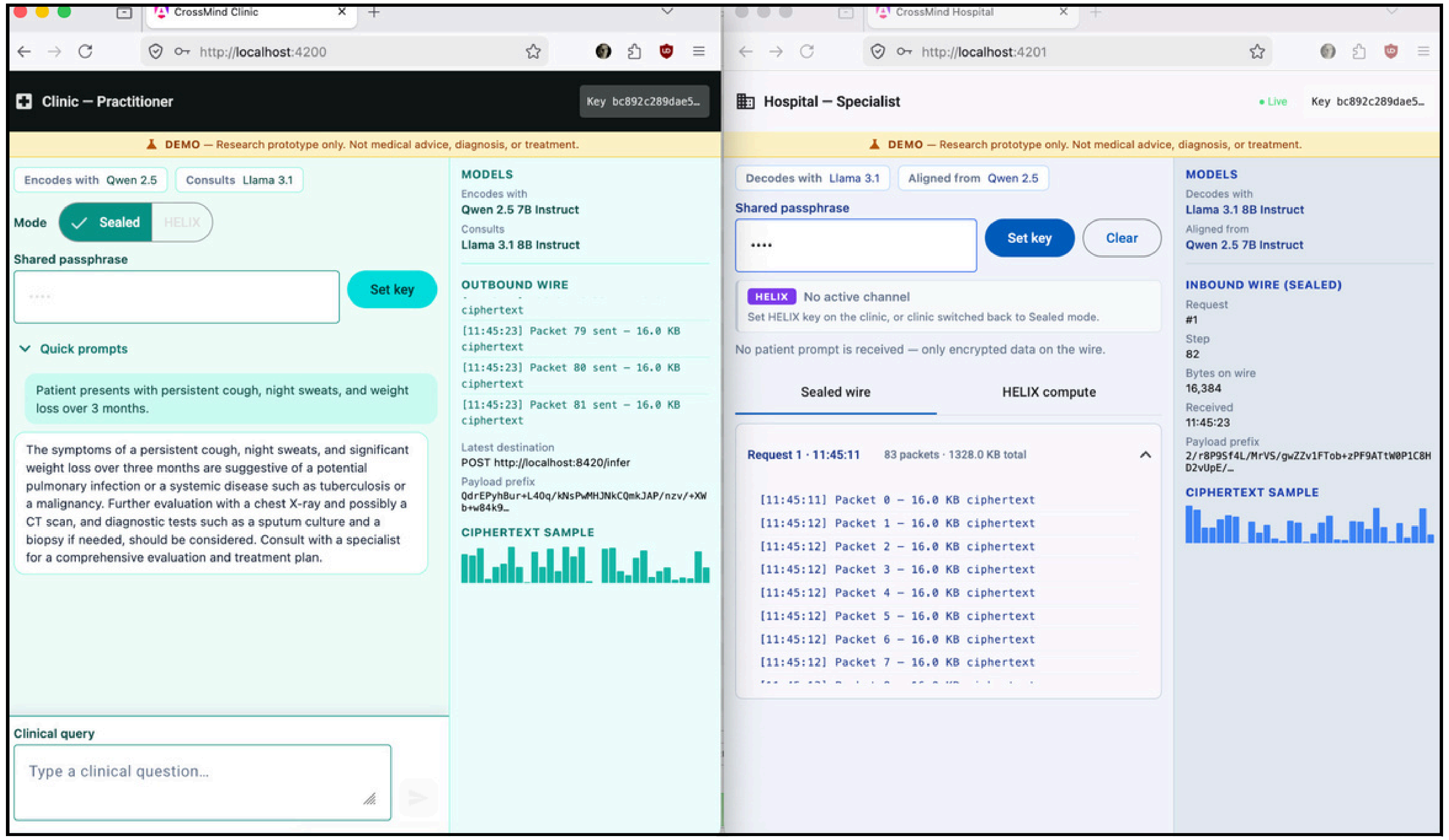
- Prompt text stays on the local machine
- Wire tap sees only rotated/encrypted vectors
- HELIX: hospital computes without seeing plaintext **(truely encrypted)**
- Department label decrypted only by clinic
- Data minimization: no full-text export needed
- Smaller remote model gets an immediate response from specialist model

What it does NOT (yet)

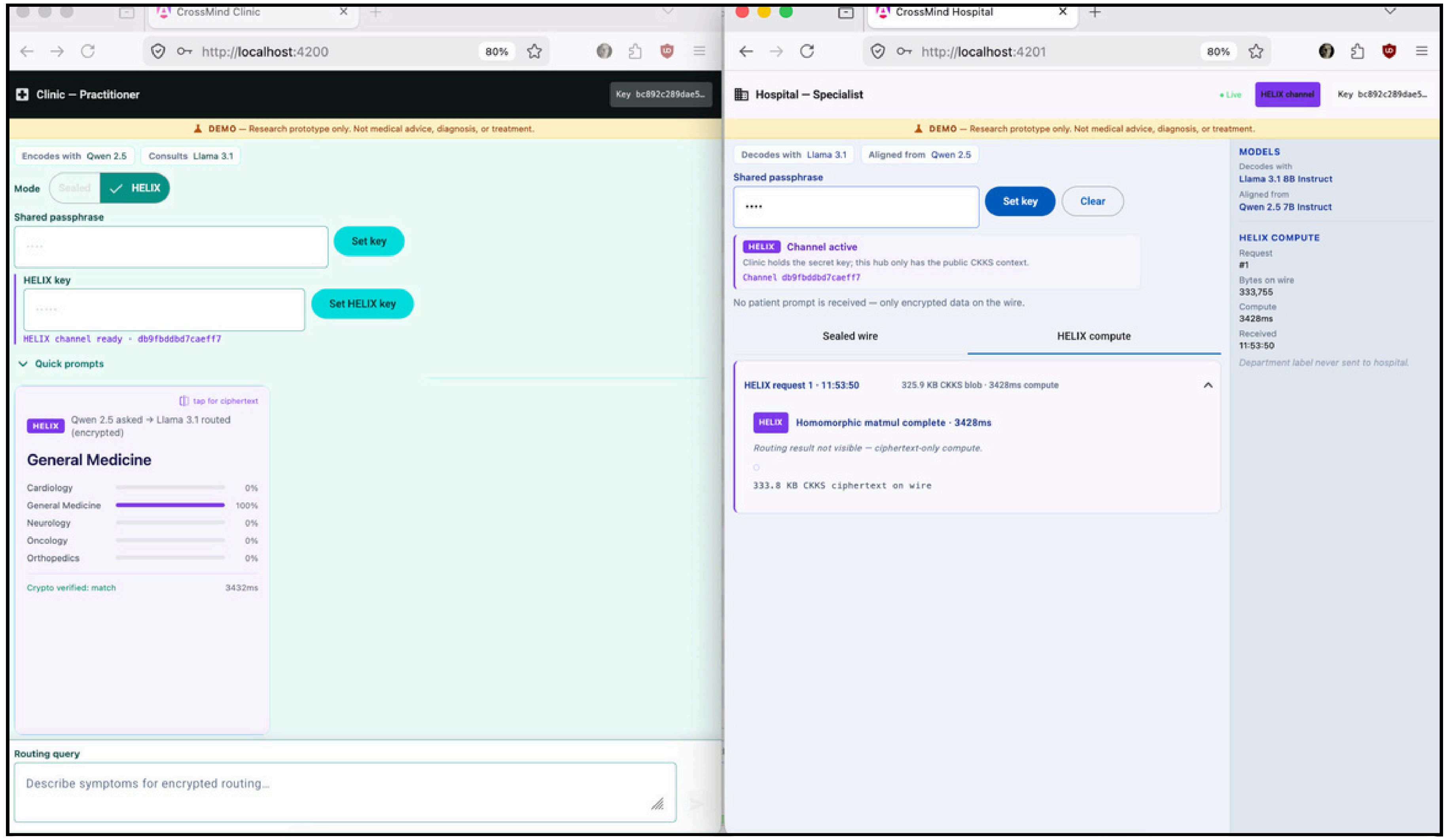
- Full encrypted generation (too expensive to compute at vocab scale today)
- Sealed generation still requires hospital-side decrypt per token
- HIPAA/GDPR compliance certification
- Guarantee hospital learns nothing from vectors alone
- Clinical-grade diagnosis or medical advice(demo project 😅)

Live demo: Clinic and Hospital connected

Working end-to-end: Clinic encodes locally (alignment cosine ~0.83) | Sealed generation streams in real-time | Hospital routes via HELIX (~3.4s/request)



Sealed mode: vectors streaming, passphrase matched



HELIX mode: encrypted routing, department decrypted by clinic only

What is to come!

- **Upload documents (PDF, TXT)**
  Send encrypted document vectors for specialist analysis without exposing content.
- **Upload images and reports**
  Medical imaging, lab reports: encoded locally, routed or analyzed remotely via encrypted vectors.
- **Multi-modal encrypted routing**
  Same HELIX pattern extended to richer inputs for better clinical results.
- **Domain-agnostic infrastructure**
  Finance, logistics, legal, industrial: wherever sensitive data needs central AI without full-text export.

Thank you

Reference: Gorbett, T., & Jana, S. (2025). Characterizing Linear Alignment Across Language Models. arXiv:2603.18908
AI BEAVERS Founder Hackathon | House of AI Hamburg | June 6, 2026